STACKS, MODULI OF CURVES: LEMMA-STABLE-REDUCTION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3245-3256 of sources/stacks-moduli-curves/pinned-original-source.tex; proof span is lines 3258-3265, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Span #43 of moduli_spans.txt (pb=398, stmt 3245-3256, proof 3258-3265); mc03 tail wave after lane-s9 handover.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0E8A: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3127–3138. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CDH: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3265–3265. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let R be a discrete valuation ring with fraction field K. Let C be a smooth projective curve over K with $K = H^0(C, \mathcal{O}_C)$ having genus $g \geq 2$. The following are equivalent*

(1) *C has semistable reduction (Semistable Reduction, Definition 0CDH), or*
(2) *there is a stable family of curves over R with generic fibre C.*

Proof. Since a stable family of curves is also prestable, it is immediate that (2) implies (1). Conversely, given a prestable family of curves over R with generic fibre C, we can contract it to a stable family of curves by Lemma 0E8A. Since the generic fibre already is stable, it does not get changed by this procedure and the proof is complete. □

Date: stacks-project:moduli-curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.